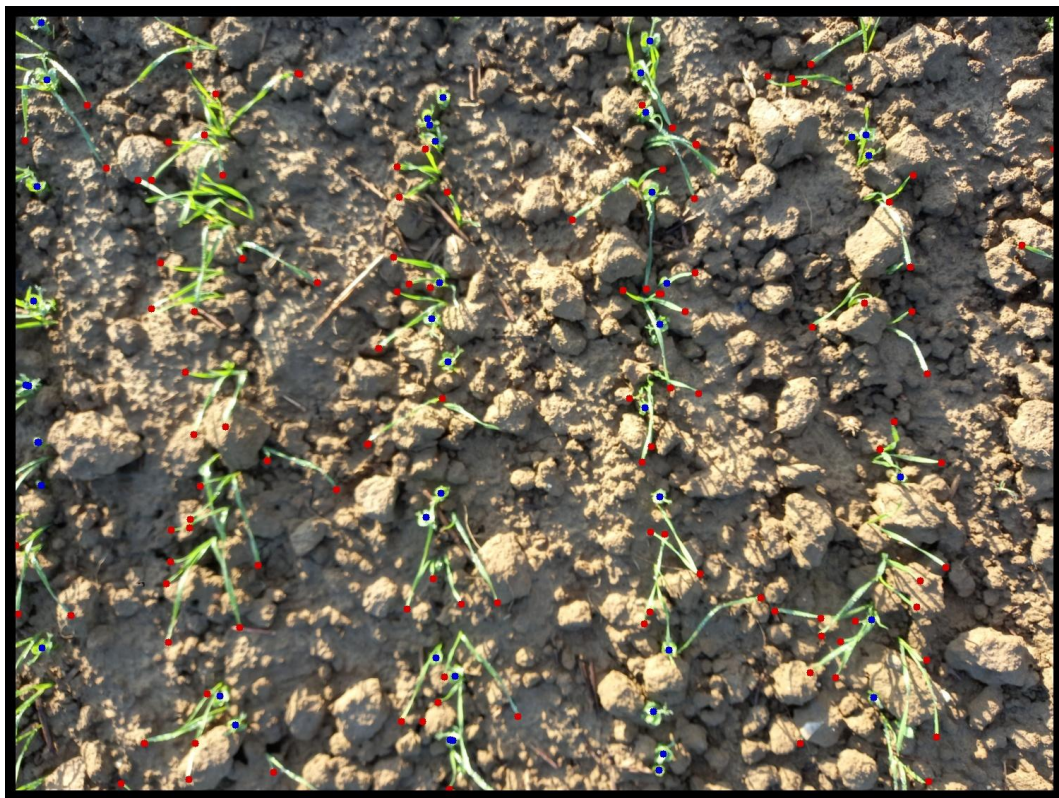


Point-Based Counting of Cereals and Legumes in Intercropped Fields

Can a crowd-counting model count crops, species by species, from a single smartphone photo?



Adapting PET to per-species plant counting



Red: wheat leaf tips · blue: pea plant centers

PET [1], a point-query transformer for crowd counting, finetuned to localize wheat and pea seedlings, tested across input resolutions and edge-border filtering.

Best test-set trade-off

Wheat

4.7% MAPE · 74% F1

1500 px, edge-border filtering on

Pea

7.8% MAPE · 86% F1

2048 px, no border filtering

MAPE alone can hide false-positive/negative cancellation: F1 is essential.

A crowd-counting model can count crops, species by species, from a single phone photo.

Come by the poster for the trade-offs, the failure cases, and what's next.



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